

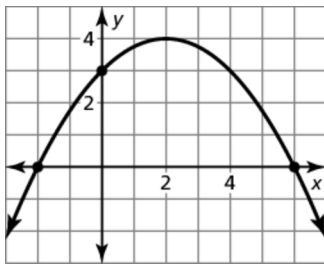
Chapter 2 Test A

1. A parabola has an axis of symmetry $x = -2$ and passes through the point $(-5, 6)$. Find another point that lies on the graph of the parabola.
2. Let the graph of g be a vertical stretch by a factor of 4 and a reflection in the x -axis of the graph of $f(x) = x^2 - 3$. Write a rule for g .
3. Let the graph of g be a translation 1 unit down and 3 units left of the graph of $f(x) = |x - 4| + 2$. Write a rule for g .
5. Explain why a quadratic function models the data. Then use a linear system to find the model.

x	1	2	3	4	5
$f(x)$	$-\frac{1}{2}$	2	10	22	38

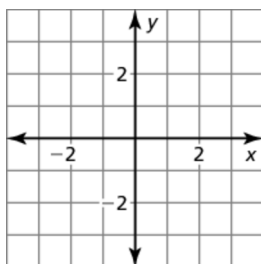
Write the equation of the parabola.

8.

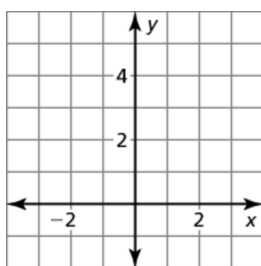


11. A bridge builder plans to construct a cable suspension bridge in your town. The cable being used will form a curve modeled by the equation $h(x) = 3x^2 - 6x + 200$, where x represents the length of cable used (in feet) and $h(x)$ represents the height of the cable (in feet). At what height will the cable hang closest to the bridge deck?

12. Graph $f(x) = 3(x + 1)^2 - 3$. Label the vertex and axis of symmetry. Describe where the function is increasing and decreasing.



13. Graph $g(x) = -2x^2 - 4x + 3$. Label the vertex and axis of symmetry. Describe where the function is increasing and decreasing.



15. The tables show the number of toy bears a toy manufacturer can sell.

Price (dollars), x	3	4	5	6	7
Bears sold (in thousands), y	84	96	100	96	84

Use quadratic regression to determine how many bears the manufacturer will sell if it charges \$9 for each bear.